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Political Polarization and International Trade: Empirical evidence from EU countries

Gao Feng
Department of International trade and business,
Shanghai Jian Qiao University
Shanghai, China
e-mail: 13057@gench.edu.cn

***Abstract*— This paper empirically examines the relationship between political polarization and international trade. The European Parliament election data are used to construct the support rate of right-wing and far right-wing parties that are then used to estimate its impact on international trade. The results show that the support rate of the right-wing parties is significantly negatively correlated with imports. The evidence we presented suggested that countries should be alert to the potential restraining effect of the rise of nationalism and anti-globalism behind the political polarization on international trade.**

Keywords: political polarization; international trade; nationalism; regression analysis

I. INTRODUCTION

Since the Battle of Seattle in 1999, the world's anti-globalization movements have never stopped, and its scale has continued to expand, with more and more countries and regions involved. Especially in recent years, with the occurrence of landmark events such as Brexit, the U.S. election, and the European refugee crisis, it can be clearly felt that non-economic factors such as nationalism and xenophobia are constantly pushing up anti-globalization sentiments, immigration conflicts. A number of trade protection measurements have been introduced since mid-2018, the most significant of which have been higher tariffs on bilateral trade between the US and China. The anti-globalization trend seems to be showing an intensifying trend.

At the same time, the development of global trade seems to have encountered the biggest crisis since the 2009 financial crisis. Figure 1 shows global trade has been exceptionally weak over the past ten years. The average annual growth rate from 2012 to 2014 was less than 2.5%. In 2015, it experienced a cliff-like decline again, with the growth rate of export trade dropping by -12.91%. Despite the trade recovery in 2017 and 2018, the overall global trade environment still tends to deteriorate, and market expectations are relatively pessimistic. So, what are the reasons behind this sluggish trade growth that has lasted for nearly 10 years? Should the anti-globalization trend of thought sweeping the world in recent years bear certain responsibility for it?

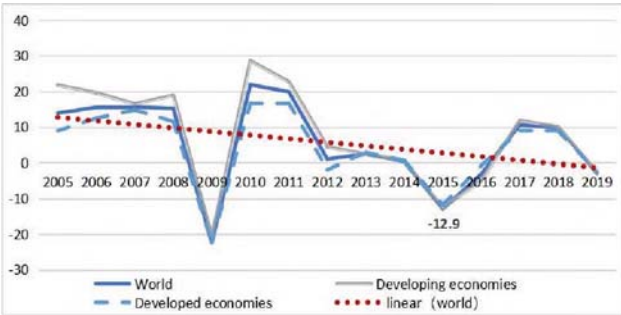

Figure.1 Average annual export growth rate, 2005-2019

Sources: UNCTAD Data Center

In this paper, we attempt to quantify the anti-globalization sentiment and construct a empirical model to answer the above question. Considering that the general rise of right-wing and extreme-right parties is the main external manifestation and political result of the rise of anti-globalization sentiments, therefore, we proxy Euro-skepticism, populism, and anti-globalization sentiments by the results of democratic elections in major European countries, more precisely, the left and right seats in the European Parliament elections, to reflect the changes in European people's attitudes towards globalization in recent years, and then to estimate its impact on international trade.

Existing relevant economic research includes, theoretically, Smith (2002) took European integration as an example, and finds that greater resistance to European integration will be faced if the region has a more deep-rooted national identity. Tabellini (2006) and Guiso et.al (2006) explained the influence of culture, values and beliefs on a country's economic system and economic growth. In terms of empirical research, Rode and Revuelta (2015) emphasize that culture affects a country's economy by influencing national policies and institutions. The empirical results show that there is a significant negative effect between populism and economic institutions. Leis and Viola (2010) find that populism seem to erodes people's beliefs and social capital, and affects the formulation of the economic system, which is also the main reason for Argentina's long-term economic stagnation and recession. Acemoglu (2010) points out that one of the biggest hidden dangers of the future development of globalization may be the increasing militarism and nationalism in various countries. Based on 20 years of transnational data, he finds that if we proxy militarism and nationalism sentiments by military spending, it can be found

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that it has a significant negative correlation with international trade. Our main contribution in this paper is to show that the left and right seats in the European Parliament elections, in our interpretation as proxy for Euro-skepticism, populism, and anti-globalization sentiments is negatively associated with trade

II. The Rise of Right-Wing Nationalism in Europe

As shown in Table 1, since the outbreak of the financial crisis in 2008, right-wing parties have begun to rise again in democracies in Europe, for example, the National Front in France, the UK Independence Party in the United Kingdom, the Five Star Movement in Italy, Alternative for Germany in German and et al. These right-wing populist parties are generally associated with positions such as neo-nationalism, anti-globalization, nativism, protectionism, opposition to immigration and European integration. Most of they believe that immigration is a negative by-product brought about by globalization, which not only causes a reduction in jobs and benefits, an increase in crime rates in the country, but also endangers national and cultural identity. Take the French National Front as an example, since its establishment, the party has been a far-right party, advocating significant cuts to immigrants and refugees from non-European countries. Its main policies also include National conservatism, French nationalism, protectionism, Euro-scepticism and anti-globalism. The National Front has opposed the European Union since its creation, and supports restricting imports from other countries through methods such as tariffs on imported goods, import quotas, and a variety of other government regulations. Although the French National Front did not ultimately win the French presidential election in 2017, it won 21.3% of the nationwide vote in the first round, and 33.9% in the 2nd round. From the numbers, what we can see is that the strength of Front National, in other words, the strength of right-wing parities has become hard to ignore.

A more representative case is that since the European Commission promulgated the refugee distribution plan in 2015, Hungary and Poland have not accepted any refugees, while the Czech Republic has accepted a total of only 12 people. In April 2018, the Hungarian ruling coalition once again won an overwhelming victory in the parliamentary elections, and Orbán was successfully re-elected as prime minister for consecutive three terms. Although Orbán did not put forward a clear election program from beginning to end before the election, and party corruption scandals continued, his speech in March made it clear that he would continue to "prevent refugees from entering Hungary" in spite of tremendous pressure from the abroad. Beyond the numbers, the messages from these elections is that the refugee crisis has further accelerated the rise of right-wing parties. There is a rise of right-wing nationalism throughout the whole Europe, even in France, which prides itself as the main promoter of European integration and Globalization. This phenomenon certainly deserves our more attention, which is the main research problem of this paper.

TABLE I. The Rise of Right-Wing Nationalism in Europe

Country	Party	Important events	Ideology
France	National Front (The National Rally after June 2018)	Won the nationwide vote in the first round of French local elections in 2015	French nationalism; National conservatism; Economic nationalism; protectionism; Right-wing populism; Anti-immigration; Euroscepticism; Anti-globalism;
U.K	UK Independence Party	Significant breakthroughs at the 2013 local elections, 2014 European elections, and 2015 general election.	Euroscepticism; Right-wing populism Economic liberalism; British nationalism National conservatism
Italy	Five Star Movement	Won the most votes of all parties for the Chamber of Deputies in 2013 general election Became the largest individual party in the Italian Parliament In the 2018 general election	Populism; Anti-globalization; Direct democracy; Soft Euroscepticism;
German	Alternative for Germany	Became the third-largest party in Germany in the 2017 federal election	German nationalism; Euroscepticism; National conservatism; Anti-Islam; Anti-immigration;
Hungary	Fidesz – Hungarian Civic Alliance	Dominate Hungarian politics on the national and local level since 2010	National conservatism; Soft Euroscepticism; Anti-immigration; Right-wing populism; Economic nationalism;

Sources: European Parliament official website

III. Data

Our measurement of trade is the Import Penetration Ratio. Import penetration rate measures the proportion of imports in a country's domestic consumption and is equal to imports/domestic demand (M/D), where domestic demand is equal to a country's GDP minus exports plus imports ie [D = GDP-X+M)]. Among them, the volume of imports and exports is from the United Nations Conference on Trade and Development UNCTAD database, and the GDP is from the International Monetary Fund IMF database.

The left and right seats in the European Parliament elections is mainly compiled from the official website of the European Parliament. The European Parliament elections began in June 1979. It is the only directly elected supranational parliament in the world. The member states of the European Union elect their members to participate in the European Parliament through

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direct universal suffrage according to the allocated quota. The term of office of parliamentarians is five years, and the latest two elections were held in 2014 and 2009 respectively. Held in the 27 member states of the European Union in 2009, the voter turnout was 42.97%, and 766 members of the European Parliament (MEPs) were elected. In 2014, it was held in the 28 member states of the European Union, with an election turnout rate of 42.61%, and a total of 751 seats were elected. It should be said that the European Parliament election involves many countries and the large number of voters, to a certain extent, it can clearly reflect the changes in the views of the vast majority of people in Europe on certain social issues.. Therefore, this paper selects the 27 countries included in both two elections as sample countries. Through text mining, we use "hard Euro-skepticism, anti-establishment, anti-immigration, and right-wing populism" in the party ideology as keywords to determine whether the parties belong to right-wing and extreme right-wing with anti-globalization sentiment, and then calculate total votes of right-wing political orientations in each country in each year. By comparing the changes of voter support for right-wing and populist parties, we get the overall disapproval for globalization and the wen can quantitatively analyze its impact on the country's international trade.For example, in 2009 and 2014 European Parliament election, as a whole, it has the following features. (Table 2)

For controlled variables, GDP per capita and the real effective exchange rate are from the UNCTAD database, while foreign direct investment and multinational company economic activities are from Organization for Economic Cooperation and Development OECD database.

TABLE II.RESULTS OF POLITICAL GROUPS IN 2009 AND 2014 EUROPEAN PARLIAMENT ELECTION

Political groups	Ideology	2014 seats	2014 seat %	2009 seats	2009 seat %	changes in seats	% change in seats
European People's Party	Centre-right Liberal conservatism Christian democracy Pro-Europeanism	221	29.43%	274	35.77%	-53	-6.34%
Progressive Alliance of Socialists and Democrats	Centre-left Social democracy Pro-Europeanism	191	25.43%	196	25.59%	-5	-0.16%
European Conservatives and Reformists Party	Anti-federalism Conservatism Economic liberalism Euroscepticism National conservatism	70	9.32%	57	7.44%	+13	+1.88%
the Alliance of Liberals and Democrats for Europe Party	Centre Liberalism	67	8.92%	83	10.83%	-16	-1.91%
European United Left–Nordic Green Left	Left-wing Democratic socialism (majority) Communism (minority) Soft Euroscepticism	52	6.92%	35	4.57%	+17	-2.35%
Greens–European Free Alliance	Green politics Minority politics Regionalism	50	6.66%	57	7.44%	-7	-0.78%
Europe of Freedom and Direct Democracy	Euroscepticism Right-wing populism	48	6.39%	31	4.05%	17	+2.34%
Non-Inscrits	/	52	6.92%	33	4.31%	+19	+2.61%
Total seats		751	100%	766	100%		

Sources: European Parliament official website

IV. MODEL AND METHODS

A. *First Difference Method*

The reason for adopting the first-order difference method for analysis is trying to capture the medium and long-term trends instead of focusing on annual fluctuations. Compared with the pooled OLS, the first difference method makes full use of the characteristics of some immeasurable factors that do not change with time, and avoids "Heterogeneity Bias".

Therefore, the model is constructed as follows:

$$\text{Import}_{it} = \beta_0 + \beta_1\text{GDPP}_{it} + \beta_2\text{LnEx}_{it} + \beta_3\text{FDI}_{it} + \beta_4\text{MultiCorp}_{it} + \beta_5\text{Unemploy}_{it} + \beta_6\,\text{Right-wing}_{it} + \varepsilon_{it}$$

$$i=1.2....N \qquad t=1.2....T \tag{1}$$

Here Import_{it} is Import Penetration Ratio of country i in year t. GDPP_{it} is GDP per capita of f country i in year t, EX_{it} is Real Effective Exchange rate of country i in year t, FDI_{it} is Inward FDI stocks in percentage of GDP of country i in year t, Unemploy_{it} represents the ratio of unemployed population to total population % of country i in year t, Multicorp_{it} represents the ratio of employees working in foreign companies to the total domestic employment of country i in year t; Right-wing_{it} represents the votes of right-wing parties in the European Parliament elections of country i in year t, ε_{it} is an error term.

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Column 1 in Table 1 reports estimates of equation (3.1). It reports an estimate of β6 = -0.4996 with standard error (SE= 0.185). This coefficient, which is statistically significant at less than five percent, implies an economically larger effect: a ten percent increase in the support rate of right-wing parties over ten years, holding all else constant, is associated with a reduction in import penetration ratio of approximately 5 percent. The empirical results basically confirms theoretical expectations. Based on the main ideologies and ruling ideas of right-wing parties, most of them are more inclined to support localism and trade protectionism, which have a certain blocking effect on imports.

B. *Dummy Variable*

The dummy variable method can not only describe and measure the influence of qualitative factors, correctly reflect the relationship between economic variables, improve the accuracy of the model, but also facilitate the processing of abnormal data. Taking into account the unique data characteristics of the right-wing support rate indicator, perhaps converting it from a continuous variable to a qualitative variable, that is, using the dummy variable method will be a simpler, more intuitive and accurate method.

Accordingly, based on the basic model (1), the model is adjusted to the following form:

$$\text{Import}_{it} = \beta_0 + \beta_1\text{GDPP}_{it} + \beta_2\text{LnEx}_{it} + \beta_3\text{FDI}_{it} + \beta_4\text{MultiCorp}_{it} + \beta_5\text{Unemploy}_{it} + \beta_6\,\text{dright} + \varepsilon_{it}$$

$$i=1.2\ldots.N \qquad t=1.2\ldots.T \tag{2}$$

Here Dright is the dummy variable of the right-wing support rate. if the right-wing support rate rose in the 2014 European Parliament election, dright=1; if not, dright=0.

Column 2 in Table 1 reports that the coefficient of the right-wing party is -0.22, and its p value is 0.000, which implies that holding all else constant, European countries with rising right-wing support rate between 2009 and 2014 will have an 22% decrease in import penetration rates than countries without rising right-wing support rate.

TABLE III. THE RIGHT-WING SUPPORT AND INTERNATIONAL TRADE, DEP.VARIABLE: IMPORT PENETRATION RATIO

	FD	Dummy
	(3.1)	(3.2)
lnGDPP_{it}	0.589** (0.231)	-0.057*** (0.019)
lnEx_{it}	-1.423*** (0.365)	0.112 (0.132)
FDIit	-0.121** (0.480)	-0.10*** (0.003)
Multicorp_{it}	0.161 (0.333)	0.683*** (0.121)
Unemploy_{it}	1.344 (0.957)	-0.418 (0.287)
right-wing_{it}	-0.4996** (0.185)	
dright		-0.223*** (0.03)

Note: The intercept term is not shown in the table. Significance level: * indicates that the significance level is 10%, ** is 5%, *** is 1%

V. CONCLUSION:

This paper found that both the first-order difference method and the dummy variable method have reached the same empirical conclusion that the support rate of the right-wing parties is significantly negatively correlated with imports. The reason may be that supporters of right-wing and far-right parties have a high probability of being nativistic and nationalistic to a certain extent, and they have certain doubts or denials about globalization. This is manifested in politics as well as economically. They are more likely to be hostile to foreign countries, foreigners, and foreign products. They support restrictions on foreign trade measures, and also support "Buy American, Hire American" by action, which may lead to a strong retrogression in the development of global trade

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